



**First Semester 2026-27
COURSE HANDOUT PART-II**

Date: 01.08.2026

In addition to part I (General handout for all courses appended to the timetable) this portion gives further details regarding the course.

Course No. : CHEM G521
Course Title : Environmental Chemistry
Instructor-in-charge : BHARTI KHUNGAR

1. Course Description:

Energy-flows and supplies, fossil fuels, nuclear energy, nuclear waste disposal, renewable energy, industrial ecology, green chemistry, ozone chemistry, effect of SO_x, NO_x as pollutants, reformulated gasoline, water pollution and treatment, organochlorine and organophosphate pesticides, eco-system effects, Toxic chemicals – Effect of dioxins, polychlorinated biphenyls (PCBs) and species of metals such as lead, mercury, cadmium etc.

2. Scope and Objective of the Course:

This course aims to provide students with a comprehensive understanding of the scientific principles underlying environmental issues and the strategies needed to achieve sustainable development. It introduces the relationship between human activities, industrial growth, energy consumption, and environmental quality, enabling students to understand both the causes of environmental degradation and the methods available to prevent or reduce it.

3. Course learning outcomes

CLO 1: Explain the principles of energy flow and evaluate conventional and renewable energy resources, including fossil fuels, nuclear energy, and their environmental impacts.

CLO 2: Analyze the concepts of industrial ecology and green chemistry for sustainable resource utilization and pollution prevention.

CLO 3: Describe ozone chemistry and assess the environmental and health effects of air pollutants such as SO_x, NO_x, and reformulated gasoline.

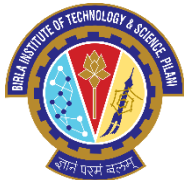
CLO 4: Evaluate the sources, effects, and treatment methods of water pollution for effective water quality management.

CLO 5: Compare the characteristics, environmental fate, and ecological impacts of organochlorine and organophosphate pesticides.

CLO 6: Assess the toxicity and environmental impacts of hazardous chemicals, including dioxins, polychlorinated biphenyls (PCBs), and heavy metals such as lead, mercury, and cadmium, and recommend appropriate pollution control measures.

4. Text Books and Reference books:

TB1. Principles of Environmental Chemistry, III Edition, James E. Girard, Jones & Bartlett Learning
RB1. Environmental Chemistry by Stanley Manahan, IX Edition, CRC Press
RB2. Environmental Chemistry by Colin Baird and Michael Cann, IV Edition, W H Freeman



5. Course Format & Weekly Workflow : 14 weeks with 3 lectures per week

Week No.	Module No.	Description	Reference	Learning outcomes
1	1	Planet Earth: Rocks, Life and Energy; The Earth's Atmosphere	TB1: 1-3	Formation of Universe, Different Layers, Formation of Oceans, Atmosphere, Ores and Metals, Origin of Life, Major Layers, Atmospheric composition, Units, Energy Balance, Particles in Atmosphere
2	2	Chemistry of Troposphere and Stratosphere	TB1: 5-6	Chemical Reactions in Atmosphere, CO, NO _x , VOCs, Automobile Pollutants, SO ₂ , Smog, Temperature Inversion, Dobson Unit, Ozone Production and Destruction, Depletion of Ozone Layer
3	3	Global Warming and Climate Change	TB1: 4	Global Temperature Changes, IR and Molecular Vibration, Residence Time, Radiative Forcing, Evidence and Effects of Global Warming
4	4	Analysis of Air and Air Pollutants,	TB1: 7	Absorption Measurements, IR spectrometry, Remote Measurements (ATMOS, LIDAR, TOMS, etc.), Automotive Emission Monitoring,
5-7	5	Water Resources, Water Pollution and Water Analysis	TB1: 8-10	Distribution of Water on Earth, Composition, Hydrologic Cycle, Unique Properties of Water, Acid Rain, Water Use and Shortage, Water Management and Conservation, Water Withdrawal, Consumption and Pollution, Types of Water Pollutant, Disease causing Agents, Oxygen consuming wastes, Plant Nutrients, Secondary Nutrients, Micronutrients, Mixed Fertilizers, Plant Nutrients and Eutrophication, Suspended Solids and Sediments, Dissolved Solids, Thermal Pollution, Sewage Treatment, Regulation of Water Quality, Sampling Methods, Types of Water Pollutants, Disease causing Agents, Oxygen consuming wastes, Plant Nutrients, Dissolved Solids, Alkalinity, Ion Chromatography, Radioactive substances.
8-9	6	Fossil Fuels, Nuclear Power and Energy Sources for the Future	TB1: 11-13	Energy Usage, Power, Energy from Fuels, Petroleum, Natural Gas, Coal, Electricity, Sequestering CO ₂ , Nuclear Energy, Energy for Future
10	7	Inorganic Chemicals in Environment Asbestos	TB1: 14, 19	Bioaccumulation of Heavy Metals, Toxic Metals, Lead, Mercury, Cadmium, Arsenic, Asbestos materials, classification, characteristics, analysis
11	8	Organic Chemicals in Environment	TB1: 15	Polychlorinated Hydrocarbons, Persistent, Bioaccumulative and Toxic Pollutants, Using EPA's PBT Profiler, Analysis of Persistent Organic Pollutants in Environmental Samples.
12	9	Insecticides, Herbicides and Insect Control	TB1: 17-18	Organophosphates, Carbemates, Transmission of Nerve Impulses, Systemic Herbicides, Analysis, Alternative Methods, Future Farming. Types,



		Toxicology		Exposure, Dose and Response, Teratogens, Mutagens and Carcinogens, Nucleic Acids, Genetic Code
13	10	Disposal of Dangerous Wastes	TB1: 20	Careless Disposal, Solid Waste, Disposal of MSW, Hazardous Waste, Policy for Management, Superfund, Radioactive Waste disposal and Regulation
14	11	Sustainability and Green Chemistry	TB1: 16	Sustainable Consumption and Production, Five Axioms, UN Commission on Sustainable Development, Life Cycle Thinking, Leapfrog Technology, Green Chemistry.

6. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Comments
Continuous Evaluation	15	15 min.	Continuous	Closed Book
Term Paper	20	-	-	Literature Survey, Seminars and Research Summaries
Mid-Semester Test	25	90 Min.	06/10 AN2	Closed Book
Comprehensive Examination	40	180 Min.	07/12 AN	Closed and Open Book

7. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

8. Consultation Hour: Friday 5.00-6.00 pm

9. Make-up Policy: Make-up would be considered only for **genuine cases** (accompanied by proof) for Mid-sem or comprehensive exam only.

10. Notices: Notices, if any, will be displayed on **Nalanda**.

BHARTI KHUNGAR
Instructor-in-charge
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